Supplementary Material for: Explainable machine learning for wheat above-ground biomass estimation under spatial cross-validation: sensor contributions and exploratory harvest forecasting

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1. Table S1: Additional vegetation indices

Table 3 of the main text lists the 21 vegetation indices (VIs), derived from Sentinel-2 (S2) and PlanetScope (PS) spectral bands, that were retained as predictors after eliminating highly correlated indices (Section 2.2.5 of the main text). Table S1 below lists the remaining 32 VIs that were part of the same 53-index base candidate set (Section 2.2.5) but were not retained in Table 3.

As described in the main text, a subset of these S2-derived indices were additionally recalculated using the Sentinel-2 Band 8A (narrow near-infrared) instead of Band 8 (broad near-infrared), following the same formula with the near-infrared (NIR) term replaced by the Band 8A reflectance; this is denoted with an asterisk (*) next to the S2 dataset code, following the same convention as Table 3 of the main text. Together, the 53 base indices (21 in Table 3 and 32 in Table S1) and their 36 Band-8A-recalculated counterparts make up the full set of 89 candidate VIs referenced in Section 2.2.5 of the main text.

Band and index notation follows Table 3 of the main text: Red, Green, Blue = visible bands; RE1, RE2, RE3 = S2 red-edge bands 5, 6, and 7; NIR = near-infrared band (S2 Band 8 / PS Band 8); SWIR1, SWIR2 = S2 shortwave-infrared bands 11 and 12; Water vapour = S2 Band 9. Formula definitions follow standard vegetation-index literature, including the references cited in Section 2.2.5 of the main text for VI selection (Peng et al., 2023; Segarra et al., 2022; Uribeetxebarria et al., 2023; Wang et al., 2022; Wu et al., 2008; Zheng et al., 2017); the exact implementation of every index listed here and in Table 3 is available in the project code repository (script `02_calcular_vis.R`).

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Table S1. Additional vegetation indices (covariates) derived from Sentinel-2 (S2) and PlanetScope (PS) spectral bands, computed as part of the 53-index base candidate set (Section 2.2.5 of the main text) but not retained in Table 3. Asterisk (*) next to the S2 dataset code indicates that the index was also recalculated using Sentinel-2 Band 8A (narrow NIR) instead of Band 8 (broad NIR).

N°	Dataset used	Name	Abbreviation	Formula
1	S2	NDVI Variant (Red-edge 3)	NDVI_2	$(RE_3 - Red)/(RE_3 + Red)$
2	S2	Red-edge NDVI (705 nm)	NDVI_{705}	$(RE_2 - RE_1)/(RE_2 + RE_1)$
3	S2*-PS	Modified Simple Ratio	MSR	$(NIR/Red - 1)/\sqrt{(NIR/Red + 1)}$
4	S2	Red-edge Modified Simple Ratio	MSR_{705}	$(RE_2/RE_1 - 1)/\sqrt{(RE_2/RE_1 + 1)}$
5	S2*-PS	Soil-Adjusted Vegetation Index	SAVI	$\frac{1.5 \cdot (NIR - Red)}{(NIR + Red + 0.5)}$
6	S2	SAVI Variant (Red-edge 3)	SAVI_2	$\frac{1.5 \cdot (RE_3 - Red)}{(RE_3 + Red + 0.5)}$
7	S2*-PS	Green SAVI	$\mathrm{SAVI}_{\mathrm{green}}$	$\frac{1.5 \cdot (NIR - Green)}{(NIR + Green + 0.5)}$
8	S2*-PS	Green OSAVI	$\mathrm{OSAVI}_{\mathrm{green}}$	$\frac{1.16 \cdot (NIR - Green)}{(NIR + Green + 0.16)}$
9	S2*-PS	Red-edge OSAVI	$\mathrm{OSAVI}_{\mathrm{rededge}}$	$\frac{1.16 \cdot (NIR - RE_1)}{(NIR + RE_1 + 0.16)}$
10	S2	-	$\mathrm{TCARI/OSAVI}_2$	$TCARI/OSAVI_2$

N°	Dataset used	Name	Abbreviation	Formula
11	S2*	SWIR 11 Related OSAVI	$\mathrm{OSAVI_{SWIR11}}$	$\frac{1.16\cdot(NIR-SWIR_1)}{(NIR+SWIR_1+0.16)}$
12	S2	SWIR 12 Related TCARI	$\mathrm{TCARI_{SWIR12}}$	$3\cdot((RE_1-SWIR_2)-0.2\cdot(RE_1-Green)\cdot(RE_1/SWIR_2))$
13	S2*	SWIR 12 Related OSAVI	$\mathrm{OSAVI_{SWIR12}}$	$\frac{1.16\cdot(NIR-SWIR_2)}{(NIR+SWIR_2+0.16)}$
14	S2*-PS	Chlorophyll Index - Green	$\mathrm{CI_{green}}$	$(NIR/Green)-1$
15	S2*-PS	Chlorophyll Index - Red-edge	$\mathrm{CI_{rededge}}$	$(NIR/RE_1)-1$
16	S2*-PS	OSAVI x Chlorophyll Index Red-edge	OSAVI x $\mathrm{CI_{rededge}}$	$\left[\frac{1.16\cdot(NIR-Red)}{(NIR+Red+0.16)}\right]\cdot[(NIR/RE_1)-1]$
17	S2*-PS	Difference Vegetation Index	DVI	$NIR-Red$
18	S2*-PS	Green DVI	$\mathrm{DVI_{green}}$	$NIR-Green$
19	S2*-PS	Red-edge DVI	$\mathrm{DVI_{rededge}}$	$NIR-RE_1$
20	S2	Two-Band Enhanced Vegetation Index	EVI2	$\frac{2.5\cdot(RE_3-Red)}{(1+RE_3+2.4\cdot Red)}$
21	S2*-PS	Red-edge Vegetation Index 1	REVI1	$NIR-RE_1$

N°	Dataset used	Name	Abbreviation	Formula
22	S2*	Red-edge Vegetation Index 2	REVI2	$NIR - RE_2$
23	S2*-PS	Weighted Difference Vegetation Index	WDVI	$NIR - 0.5 \cdot Red$
24	S2-PS	Green-Red Vegetation Index	GRVI	$(Green - Red)/(Green + Red)$
25	S2*-PS	Green Ratio Vegetation Index	GRVI2	$NIR/Green$
26	S2*-PS	Green Atmospherically Resistant Index	GARI	$\frac{NIR-(Green-1.7\cdot(Blue-Red))}{NIR+(Green-1.7\cdot(Blue-Red))}$
27	S2*-PS	Leaf Chlorophyll Index	LCI	$(NIR - RE_1)/(NIR - Red)$
28	S2	MERIS Terrestrial Chlorophyll Index	MTCI	$(RE_2 - RE_1)/(RE_1 - Red)$
29	S2*	Normalized Difference Red-edge Index 2	NDRE2	$(NIR - RE_2)/(NIR + RE_2)$

N°	Dataset used	Name	Abbreviation	Formula
30	S2	Normalized Difference Red-edge Index (RE3/Green)	NDREI	$(RE_3 - Green)/(RE_3 + Green)$
31	S2*	Normalized Difference Moisture Index	NDMI	$(NIR - SWIR_1)/(NIR + SWIR_1)$
32	S2*-PS	Kernel NDVI	KNDVI	$\frac{1 - \exp\left(-\frac{(NIR - Red)^2}{2 \cdot 0.15^2}\right)}{1 + \exp\left(-\frac{(NIR - Red)^2}{2 \cdot 0.15^2}\right)}$

2. Equations S1-S3: Performance metrics

Section 2.3.6 of the main text reports model performance using the mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2), defined in Equations S1-S3 below, where y_i and $\hat{y}_i$ are the observed and predicted AGB values for sample i, $\bar{y}$ is the mean observed AGB across the n samples in the evaluation set, and n is the number of samples.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{S1}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{S2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{S3}$$

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